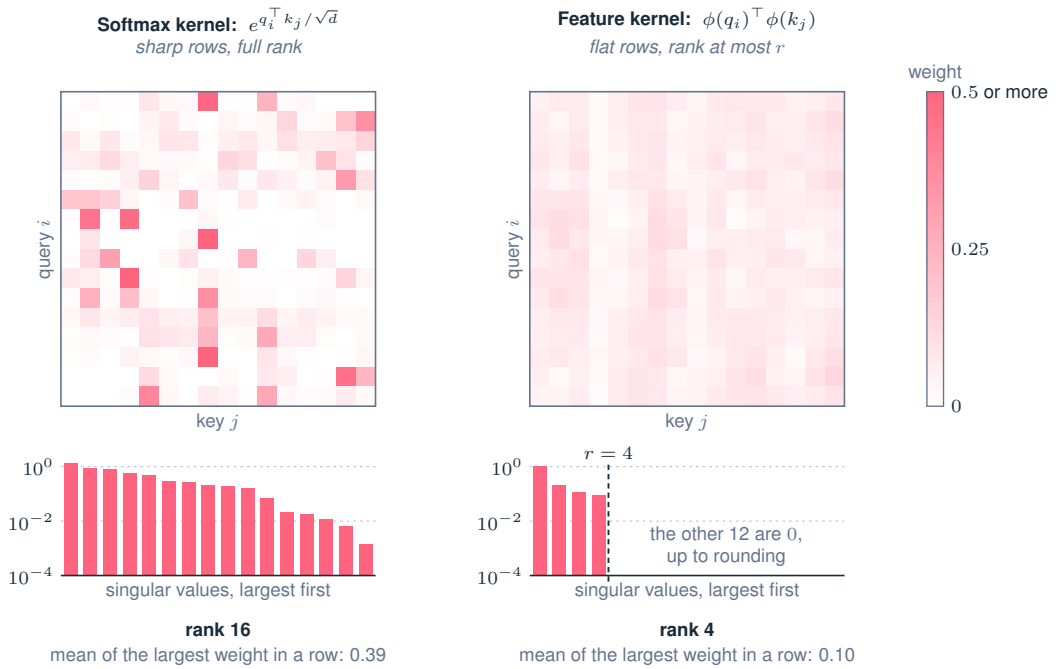


rank 16

mean of the largest weight in a row: 0.39

rank 4

mean of the largest weight in a row: 0.10

The same queries and keys in both panels: $n = 16$ tokens, $d = 4$, entries of Q and K independent $N(0, 1.5^2)$, no mask. Linear attention uses $\phi(x) = \mathrm{elu}(x) + 1$, so $r = d = 4$. A uniform row would put $1/16 = 0.0625$ on each key.